

Rational Points on Elliptic Curves Chapter 3

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1 Introduction

The set of rational points on C is finitely generated. Let $Q_1, Q_2, \dots, Q_n$ be coset representatives. Pick any $P_1 \in C(\mathbb{Q})$. By construction there exists Q_{a_1} such that $P_1 - Q_{a_1} = 2P_2 \in 2\Gamma$. We have

$$4h(P_2) - \kappa \leq h(2P_2) = h(P_1 - Q_{a_1}) \leq 2h(P_1) + \kappa_{a_1},$$

where κ_{a_1} is the κ_0 associated with the inequality $h(P + P_0) \leq 2h(P) + \kappa_0$ for $P_0 = Q_i$. Hence, $h(P_2) = \frac{1}{2}h(P_1) + \frac{\kappa' + \kappa}{4}$, where $\kappa' = \max \kappa_i$. If moreover $h(P_1) \leq \kappa' + \kappa$, we have $h(P_2) \leq \frac{3}{4}h(P_1)$.

We can repeat this process to P_2 , producing points $P_1, P_2, P_3, \dots, P_m$, where m is minimal such that $h(P_m) \leq \kappa' + \kappa$. This process terminates since $h(P_i) \leq (3/4)^{i-1}h(P_1)$ and i can be chosen to be sufficiently large.

Substituting backwards, we find

$$P_1 = Q_{a_1} + 2Q_{a_2} + 4Q_{a_3} + \dots + 2^{m-2}Q_{a_{m-1}} + 2^{m-1}P_m,$$

where $h(P_m) \leq \kappa' + \kappa$. This means

$$C(\mathbb{Q}) = \langle Q_1, Q_2, \dots, Q_n \rangle \cup \{P \in C(\mathbb{Q}) : h(P) \leq \kappa' + \kappa\}.$$

The set on the right is finite by the above lemmas, so $C(\mathbb{Q})$ is finitely generated.

Lemma: Let $\phi, \psi \in \mathbb{Z}[X]$ be polynomials with no common roots and $\phi(0), \psi(0) \neq 0$. Let the degrees of ϕ, ψ be e, f , respectively. There exists an integer R such that $g = \gcd(n^e \phi(m/n), n^f \psi(m/n)) \mid R$ for all $m/n \in \mathbb{Q}$ in simplest form.

Proof. There exist $u, v \in \mathbb{Q}[X]$ such that $u(X)\phi(X) + v(X)\psi(X) = 1$. Let $\Phi(m, n) = n^e \phi(m/n)$ and $\Psi(m, n) = n^f \psi(m/n)$ be homogenous polynomials in m, n . Plugging n/m in (there is only one such case where $m = 0$), there exist homogenous U, V and integers N, k such that $U\Phi + V\Psi = Nn^k$. Hence $g \mid Nn^k$. $f(X) = X^e \phi(1/X)$ and $g(X) = X^f \psi(1/X)$ share no roots and similarly, $g \mid Mm^j$ for some M, j . $g \mid \gcd(Nn^k, Mm^j) \mid \gcd(Nn^k, M) \gcd(Nn^k, m)^j \mid MN^j$.

Lemma: Let $\phi, \psi \in \mathbb{Z}[X]$ have no common roots. Let the maximum of the degrees of ϕ, ψ be d . Then there exists R such that $g = \gcd(n^d \phi(m/n), n^d \psi(m/n)) \mid R$ for all simplified m/n .

Proof. WLOG $\deg \phi \geq \deg \psi$. We have $g = \gcd(am^e + \dots, bm^f n^h + \dots)$. $\gcd(n, n^d \phi(m/n)) = \gcd(n, am^e) = \gcd(n, a)$. Hence $g \mid a^h \gcd(am^e + \dots, bm^f + \dots)$.

Let $a(X), b(X)$ denote the dehomogenized polynomials of $am^e + \dots$ and $bm^f + \dots$, resp and A, B the homogenous versions. Sps at least one of $a(X), b(X)$ vanishes at 0. WLOG $a(X)$ vanishes at 0. Then $b(X)$ can't vanish at 0. $A = am^e + \dots + lm^v n^?$ and $B = bm^f + \dots + cn^f$. $\gcd(m^v, B) \mid \gcd(m, B)^v \mid c^v$. Then apply the previous lemma to $am^e + \dots + ln^?$, B AWD.